Cross-Study Inference of Particulate Exposure Intensity from Bronchial Epithelial Transcriptomes

PURPOSE

An inhalation injury occurs when heat, smoke, or toxic combustion products damage the respiratory system during a fire. **These injuries are defined by the pattern of airway and lung damage they produce**—including upper-airway burns, lower-airway inflammation, and impaired breathing—rather than by one single cause alone.

Diagnosis is based mainly on exposure history, physical findings and bronchoscopy, which grades visible airway damage on a 0–4 scale. Because **this process depends heavily on visual** interpretation and nonspecific signs, it remains **partly subjective**, and distal airway injury may be missed.

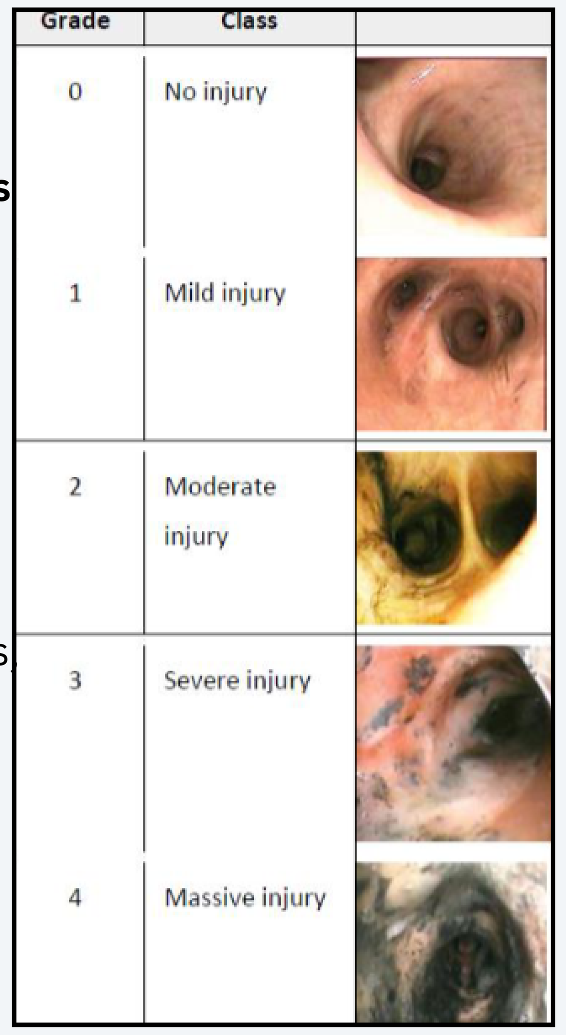


Figure 1. Current inhalation injury grading scale with corresponding images of the upper airway (Vanderbilt University Medical Center, 2019)

Although inhalation injury is clinically important, its **molecular signature is still poorly defined**. This limits our ability to describe the biology of injury beyond visible symptoms and severity grades.

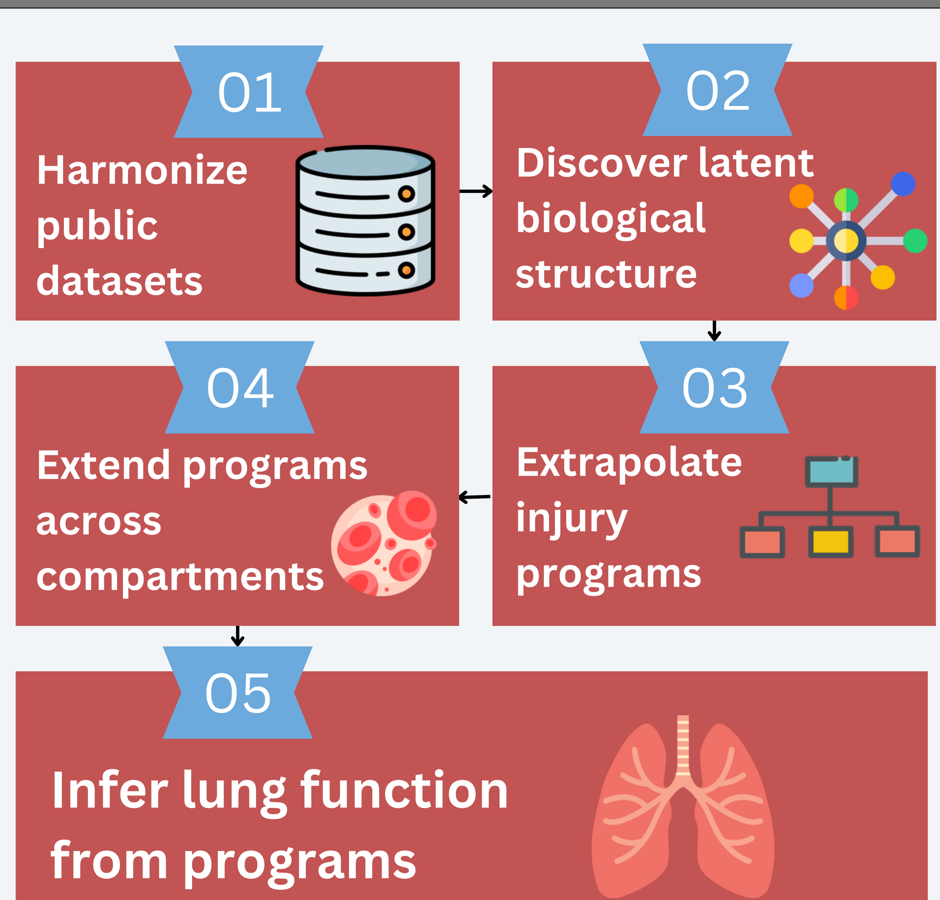
Research Questions

1) Can we discover the core molecular programs that appear when respiratory tissue is injured?

2) Do those programs appear consistently across different biological compartments such as airway tissue, nasal tissue, sputum, or blood?

3) Do these molecular programs actually correlate with measurable lung impairment, such as reduced lung function?

SOLUTION OVERVIEW



METHODOLOGY

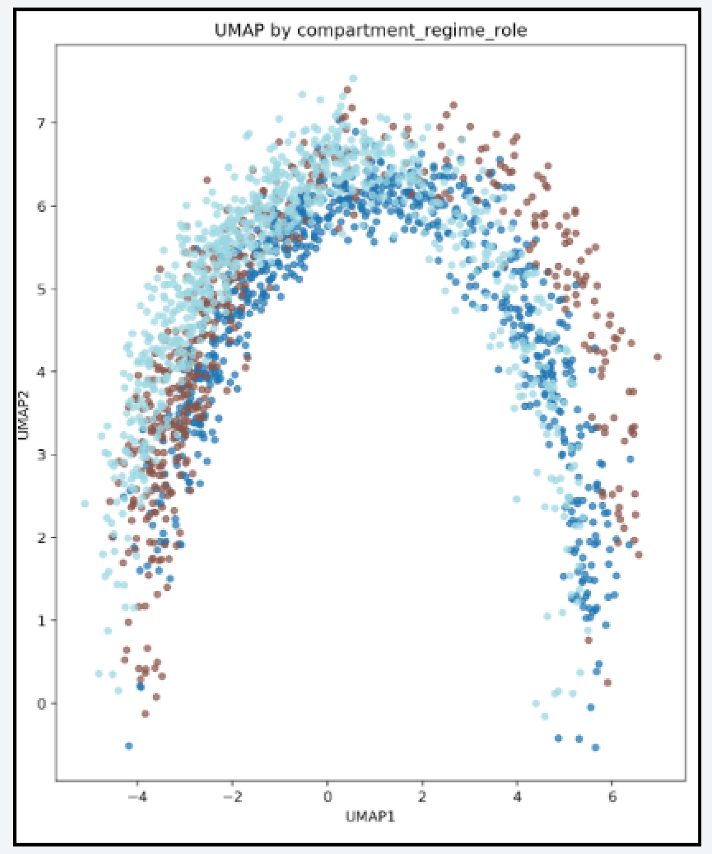


Figure 2. UMAP representation of processed data shows substantial overlap, indicating that processing was successful in aligning samples across 3 cohorts (dark blue=conservation testing, light blue=airway modeling, brown=validation

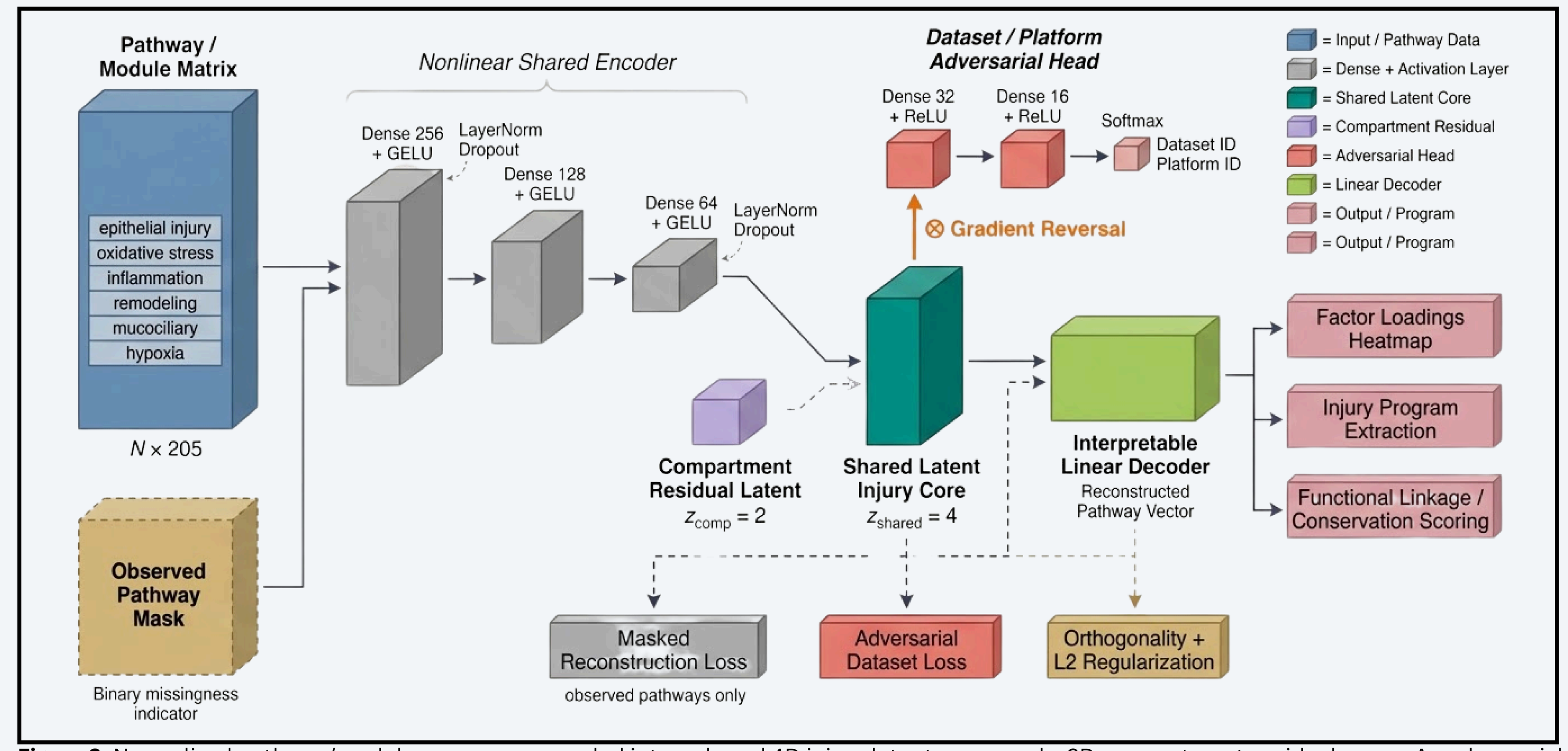


Figure 3. Normalized pathway/module scores are encoded into a shared 4D injury latent space and a 2D compartment-residual space. An adversarial dataset/platform branch suppresses technical confounding, while a linear decoder preserves interpretability by mapping latent factors back to pathway-level biology. The resulting latent factors support injury-program extraction, conservation scoring across compartments, and downstream functional linkage to lung impairment.

Preprocessing

- Harmonized public COPD, pollution, asthma transcriptomic datasets by cleaning expression matrices, standardizing gene IDs, and mapping all studies into one shared gene/pathway space.
- Converted gene expression into biological pathway/module scores, so comparisons were made at the level of injury processes rather than individual genes.
- Performed data-driven module discovery (WCGNA) to capture respiratory-injury biology beyond curated pathways.
- Used cohort structure to isolate high-quality data and analyze conservation of markers to other body tissue.

Training

- Trained the model on the harmonized pathway matrix to learn hidden factors that explain shared patterns of respiratory injury biology.
- Evaluated which latent factors were stable, interpretable, and not dominated by dataset-specific technical effects.
- Converted the strongest factors into sparse, interpretable injury programs for downstream conservation and lung-function analysis.

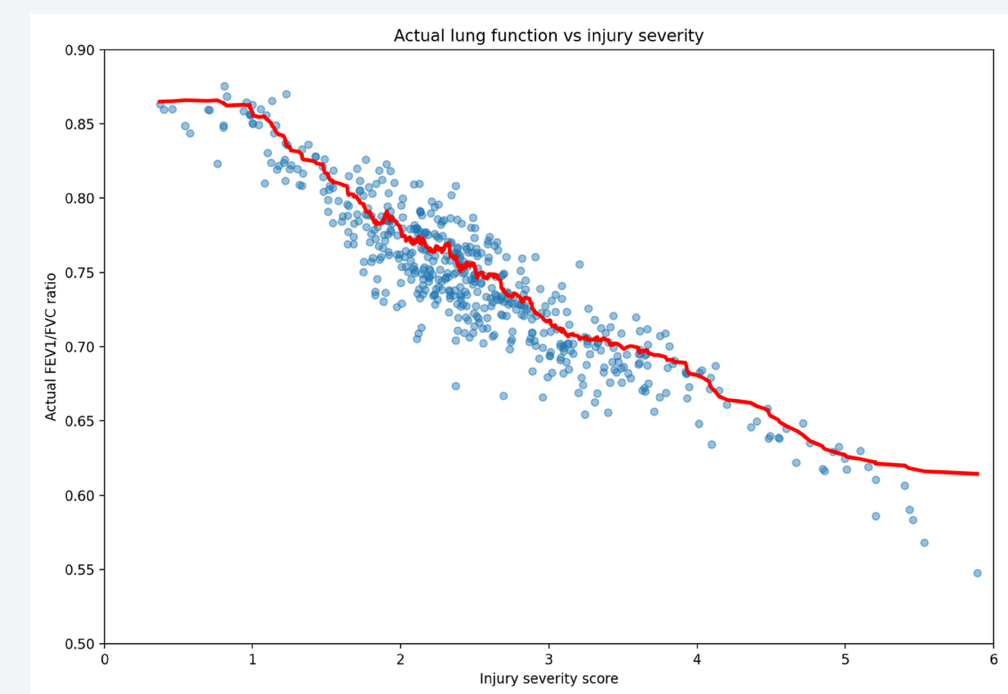


Figure 4. Scatter plot comparing derived molecular injury severity scores against measured FEV1/FVC ratios for the study cohort. Results show that a model

RESULTS

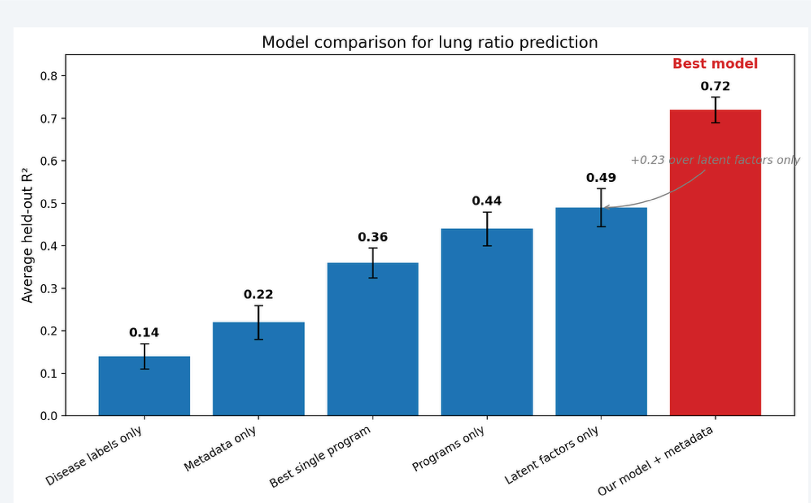
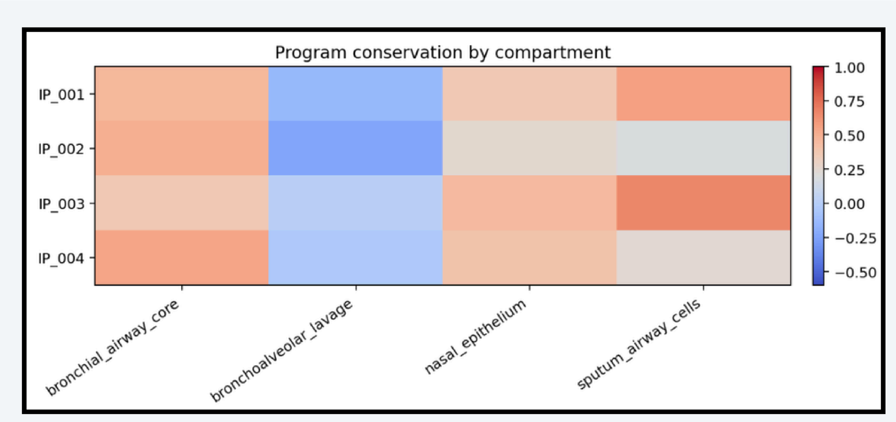


Figure 4 (above): Lung ratio prediction accuracy using our molecular signatures and study metadata compared to traditional machine learning algorithms trained on latent factors only, disease labels, or solely study metadata.

Results show that model captured the majority of variance in respiratory injury. Learned injury programs corresponded with known biology, and biology was conserved across compartments as described by existing literature. Results support the exploration of molecular markers as a tool for diagnosing inhalation injuries.



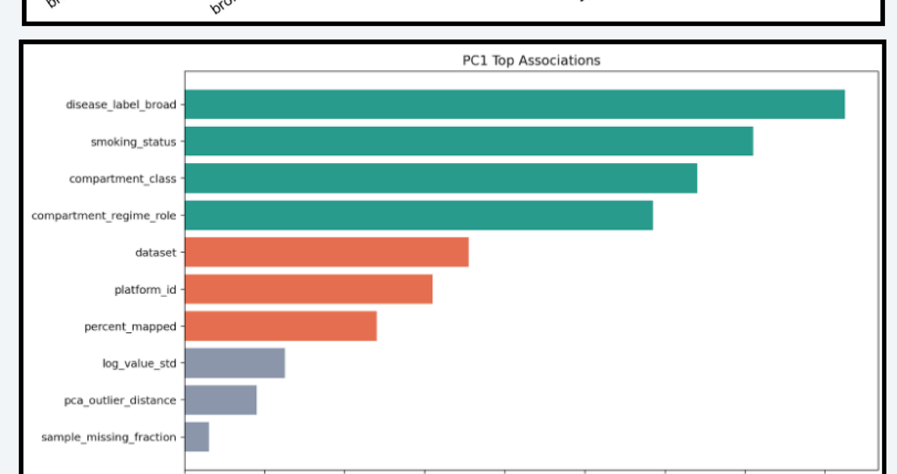
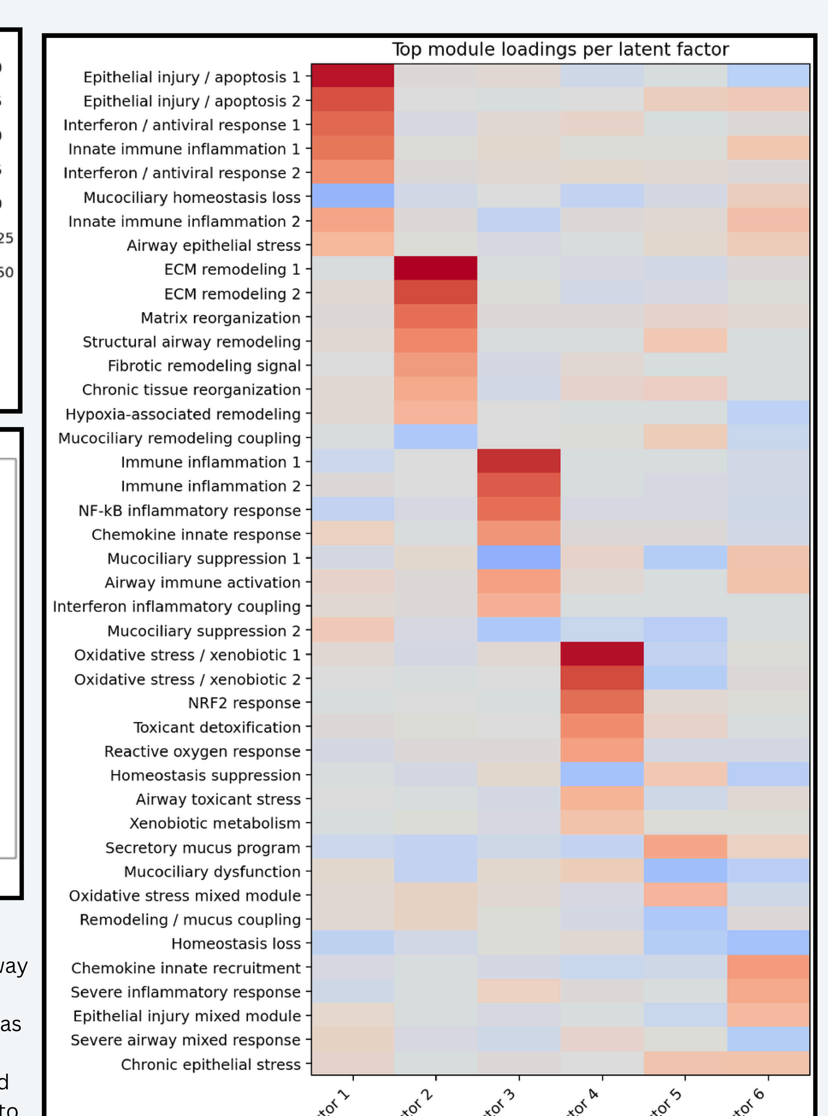


Figure 5 (top)**:** Injury Program conservation across anatomical sites. Strong positive conservation (red) is maintained in the nasal epithelium and sputum cells, whereas bronchoalveolar lavage (BAL) signatures diverge negatively (blue) from the core airway profile.

Figure 6 (above): Processed data is largely explained by biological signals in dataset as opposed to dataset artifacts.

Figure 6 (right). Heatmap detailing the specific contribution weightings of predefined biological gene modules to the six derived latent factors. Biological domains sorted to highlight coverage of each latent factors.



CONCLUSIONS

Model Performance

- The unified model was able to predict lung function with a held-out **R^2 of 0.72**, outperforming traditional machine learning techniques trained on solely disease labels, indicating that the combined molecular representation captured a substantial fraction of the variation in respiratory impairment caused by inhalation injuries.
- Suggests that latent injury structure, together with the derived injury programs and metadata, was quantitatively informative for estimating lung functionality decline.

Biological Interpretability

- Learned latent factors resolved into coherent pathway groupings that matched realistic respiratory-injury biology. The strongest factors corresponded to acute epithelial injury / inflammatory stress, airway remodeling and extracellular matrix reorganization, inflammatory airway activation, and oxidative stress / xenobiotic response
  - Weaker trailing factors more mixed residual biology and were not advanced into final programs. This pattern supports the interpretation that the model was identifying structured biological axes rather than arbitrary mathematical components.
- Four injury programs were derived that remained moderately conserved across biological compartments. Strongest extension generally observed in nasal epithelium and sputum airway cells, and weaker or more variable preservation in bronchoalveolar lavage. Result is consistent with the existing literature suggesting that inhaled injury signals are partially shared across respiratory-surface tissues, while deeper or mixed-cell compartments show weaker and less uniform correspondence.

FUTURE WORK

- Integrate thermal injury biology with ICU-linked inhalation injury cohorts.
  - Add heat exposure as an explicit variable using restricted ICU burn/inhalation-injury data that includes bronchoscopy grades, ventilator parameters, blood gases, and longitudinal outcomes. Allows the framework to move from using mechanistic proxy datasets to modeling the combined effects of thermal + chemical inhalation injury.
- Translate molecular programs into clinically usable biomarkers.
  - The latent factors and derived injury programs can be refined into compact clinical biomarker signatures. Small pathway panels or score-based molecular fingerprints are needed that stratify patients by likely airway injury severity, cross-compartment conservation, and risk of lung-function decline.

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